

Laboratory Test and Selection of Ultrasonic Sensors for Ultrasonic Cane for Blind People

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Abstract— The proposed study presents an analysis of the methods and means of detecting objects using an ultrasonic sensor for the needs of the ultrasonic cane control system for blind people. An Arduino-based module was developed and tested in laboratory and field conditions. The obtained results support the thesis that ultrasound scanning of moving objects allows avoiding existing obstacles. The results support the thesis that when measuring the distance to an object with the HC-SR04P ultrasonic sensor at a temperature of -7 °C, the absolute error is 2.64%, and at a room temperature of 24 °C, the absolute error is 0.96%, which proves that the temperature change in the environment can affect the measured actual distance to the object.

Keywords- ultrasonic; cane; electronic control system; blind people.

I. INTRODUCTION

In recent years, technology has entered every aspect of our daily lives. Technological development in almost every scientific field has led to a significant improvement in the quality of life [1], [2].

The complete analysis of existing electronic systems covers an extensive research in the scientific literature of various and many types of devices, patents and products for the visually impaired. Understanding where these systems focus and how they interact with the user to restore or replace as much as possible their vision gives insight into how these people live and how important this is to society. The state and the scientific community need to work as a team and mobilize, so that all together contribute to helping people with disabilities live as normally and fully as possible [2], [3], [4].

From the processed information and the performed analysis, it was established [5] that there is a lack of reliable statistical information on the number of persons with visual impairments in Bulgaria, which hinders the formation and implementation of adequate state policies in this area. There are no comprehensively written regulations, documents and a built-up base for technical methods, means and standards for dealing with visual disorders through promotion, prevention, treatment and rehabilitation. In the event of a disease [6] leading to total or partial visual impairment, the individual needs to use different compensation pathways.

After the analysis and evaluation of existing patents and methods [2], [7], [8] for building sticks for blind people, a concept for building a stick and the necessary element base was chosen.

The purpose of the proposed research is to present an analysis of laboratory test results and selection of ultrasonic sensors for an ultrasonic cane for blind people to achieve optimal field of view and performance.

II. ANALYSIS OF VISUAL FIELD ASSESSMENT METHODS AND TOOLS

A. Visual field research methods

The summary of the most significant smart sticks for visually impaired people presented in the international scientific literature [2], [9], [10], [11] showed that the minimization of the control electronic system determines the use of a microcontroller and as a mandatory element - ultrasonic sensors included in various configurations. In order to connect an electronic system to the natural rhythm of a blind person, it is necessary to first study the field of view of a sighted person and to study and configure on this basis the field of view of the ultrasonic sensors and their placement.

The space that a person can see with fixed gaze is the visual field. The human eye contains millions of cells. The cells that help with vision are of two types - rods and cones. The cones are sensitive to color and work well when there is a lot of light, and make up central vision. Peripheral vision uses rods and these cells are good at picking up any movement. Good peripheral vision shows people when another object is near them or will soon be, helping to prevent accidents and collisions [12].

When the eye fixes only one point, and a person's head and body are motionless, the part of the perceived surrounding space is significantly limited only to the field of view. The central part of the visual field occupies 1...2° around the fixation point. The rest of the visual field is limited by the formations of the facial skull and refers to the peripheral part of the visual field. Within these boundaries, the visual field has blind "islands" of scotomas. The so-called physiological scotoma corresponds to the optic nerve papilla [12]. The outer periphery of the retina perceives only white color, where in the corresponding direction the eye perceives the space as follows: temporally 90°, nasally – up to 60°, above – up to 60° and below – up to 70°, Fig. 1.

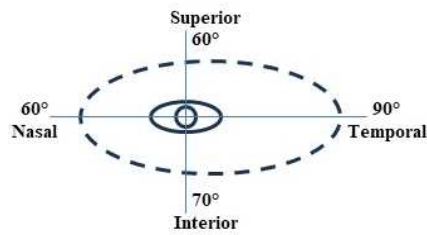


Figure 1. Limits of peripheral vision in normal.

The examination of the visual field is carried out in the four main meridians (upper, lower, external and internal) and in the four hairs, i.e. the limits of the field of vision are defined in a minimum of 8 meridians.

The methods for examining the visual field can basically be divided into two groups: perimetric and campimetric. In perimetric methods, the hollow hemisphere of the eye (the retina) is projected onto a similar hollow hemisphere - a screen. This is done with the perimeter of Forster, Madjore, Kugel, etc. When the screen on which the visual field is examined is flat, the methodology is called campimetry [12].

Forster's perimeter is a quarter-arc or semi-arc, which in its rotation along all meridians practically realizes a hemispherical screen for examination. During the examination, in addition to determining the boundaries of the visual field, scotomas are also sought in it [12]. This type of perimetry is also called kinetic because the object is moving. In contrast to it, there is also a methodology in which the visual field is examined with a fixed mark with changing parameters. This is the so-called static perimetry.

The campimeter is a flat screen measuring 2 x 2 m. Campimetry is conducted from different distances, with the sizes of the scotomas varying. It is used in expert practice to detect simulations and aggravations. An inconvenience when working with the campimeter is that it can be used to examine only the more central areas of the visual field - about 30° [12]. However, an advantage of the method (over perimetry) is that campimetry can detect small paracentral scotomas, difficult to detect and often missed by perimetry.

Based on the study of the field of vision of a sighted person, the following hypothesis was raised: In the horizontal plane, the eye sees up to 150°, and in the vertical plane - up to 130°. Based on this hypothesis, the following ranges for the field of view of the ultrasonic sensors are set for research: horizontal plane – up to 160°, vertical plane – up to 140°, which includes the present human field of view, Fig. 2.

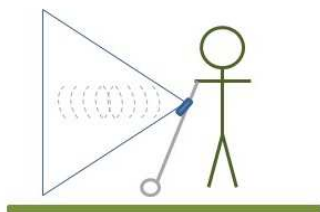


Figure 2. Perimeter of operation of the ultrasonic probe, covering the vertical plane.

B. Examination of the location of the ultrasonic sensors in the horizontal plane

The study of the location of the ultrasonic sensors in the horizontal plane of the field of view is carried out in laboratory conditions with sensors of different manufacturers, Table 1.

TABLE 1. SELECTION OF SENSORS

Manufacturer	Sensor name	Measuring angle
ITEAD – China	HC-SR04	43.2°
SparkFun Electronics	HC-SR04P	42.8°

In order to check the measurement accuracy of the sensors of the two manufacturers, a test comparative measurement was made. It is found that there is no deviation in the distance measurement, but there is a 0.4° difference in the measurement angle. Based on this study, an ultrasonic sensor HC-SR04P was selected, which has a coverage angle of 42.8°, Fig. 3.

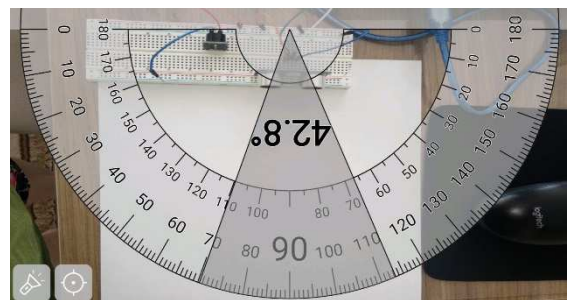


Figure 3. Coverage angle of an ultrasonic sensor sensor.

Based on the accepted hypothesis of a horizontal plane of the field of view with a coverage angle of up to 160°, three ultrasonic sensors were selected, located as shown in Fig. 4, and defining an angle of the field of view of 128.4°.

If a fourth sensor were added to the sensor arrangement of Fig. 4, this would result in a wider field with a lot of information loss due to the fact that the areas overlap.

As a next step in the study of the location of the Ultrasonic sensors, the angle of the location of the sensors relative to each other was tested, Fig. 5. When conducting the experiment, the location of the three fields of view of the Ultrasonic sensors was decisive to achieve the total field of view of the device up to 160°, Fig. 6, without overlapping of the individual zones.



Figure 4. Angle of coverage of three ultrasonic sensors.

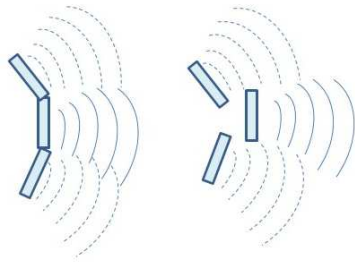


Figure 5. Location of the three ultrasound sensors - fields of view.

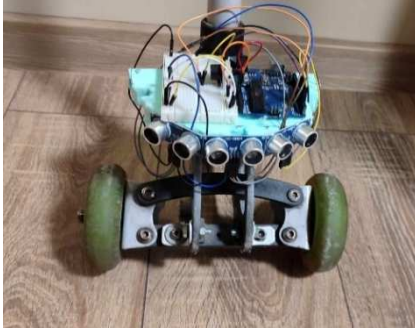


Figure 6. Prototype of the stick with the ultrasound sensors in place.

C. Testing the ultrasonic cane control system

Based on the accepted hypothesis of a horizontal plane of the field of view with a coverage angle of up to 160° and the selected placed US sensors on the stick for blind people, Fig. 6, a study was carried out in laboratory and field conditions, choosing different distances: short (20 cm), medium (50 cm) and long (80 cm). If there is an object in the range of the sensor, the distance to the detected object is calculated and signaled by sound and vibration signaling. If there is an object less than 50 cm away. the buzzer plays a sound signal that increases in frequency as the distance between the object and the blind person decreases.

Table 2 and Fig. 7 present the results of testing the ultrasonic system in laboratory conditions at different distances at an ambient temperature of 24 °C.

TABLE 2. ULTRASOUND SYSTEM TEST IN LABORATORY CONDITIONS

Set Distance, [cm]	Measured Distance, [cm]	Absolute Error, [%]
20	20.36	1.80%
30	30.68	2.26%
40	40.34	0.85%
50	50.10	0.20%
60	60.30	0.50%
70	70.78	1.11%
80	80.45	0.56%
90	90.84	0.93%
100	100.45	0.45%
Average Absolute Error		0.96 %

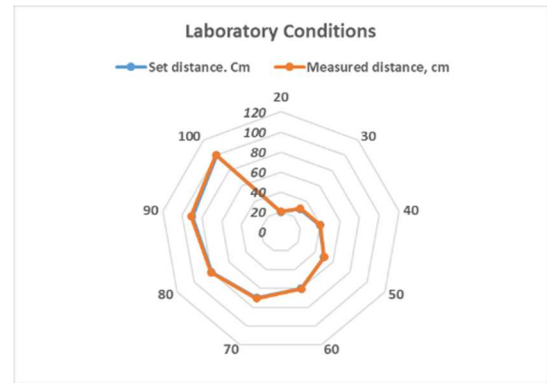


Figure 7. Laboratory Conditions.

Table 3 and Fig. 8 present the results of testing the ultrasonic system in field conditions at an ambient temperature of -7 °C at different distances.

TABLE 3. ULTRASOUND SYSTEM TEST IN FIELD CONDITIONS

Set Distance, [cm]	Measured Distance, [cm]	Absolute Error, [%]
20	20.53	2.65%
30	30.87	2.90%
40	40.90	2.25%
50	51.16	2.32%
60	61.35	2.25%
70	71.74	2.48%
80	82.10	2.62%
90	92.85	3.17%
100	103.16	3.16%
Average Absolute Error		2.64%

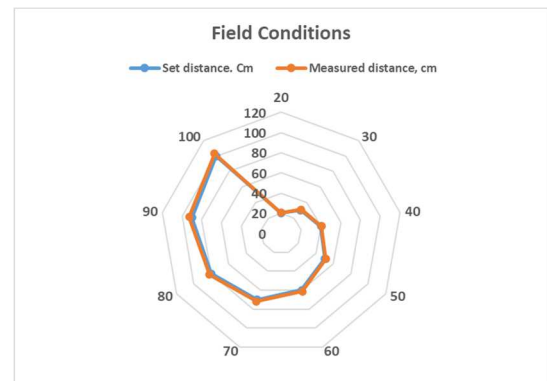


Figure 8. Field Conditions.

From the measurements made in laboratory and field conditions, it is clear that under field conditions the average absolute error increases by 1.6%. A 180° field of view angle is achieved under both laboratory and field conditions.

III. CONCLUSION

The proposed signaling method (20, 50, 80 cm) only reports the distance and not the direction of the obstacle. The direction of the obstacle is part of the next study, related to the undercarriage at the base of the stick control. The undercarriage part of the stick consists of motors that will correct the direction of the stick when sensors detect obstacles. Engineering solutions for ultrasonic cane control system for blind people in Arduino based environment are proposed. Ultrasonic cane control system has been tested in laboratory conditions, and the obtained results support the thesis that the ultrasonic scanning of the course of movement of a blind person allows to avoid existing obstacles.

The findings substantiate the hypothesis positing that variations in environmental temperature exert discernible effects on the precision of distance measurements to objects when employing the HC-SR04P ultrasonic sensor. Specifically, at a temperature of -7 °C, the absolute error in distance determination is quantified at 2.64%, whereas under ambient conditions of 24 °C, the absolute error diminishes to 0.96%. This empirical evidence convincingly demonstrates the influence of environmental temperature fluctuations on the accurate determination of the object's actual distance.

To achieve adaptive solutions for visually impaired individuals, the results obtained not only enrich our current understanding but also open opportunities for future research, namely through the development of robust methodologies and technological improvements.

ACKNOWLEDGMENT

The article presents results from Project 2023–FEEA-02, with the financial support of the National Science Fund of the University of Ruse.

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